One story, seven minutes

You've done three sections. They're not seven separate topics; they're one story with seven scenes. Learn the story, and the facts hang off it.

THE STORY (memorize this paragraph; everything else is detail):
Space gives you nothing for free. **Getting there** is falling and missing: go sideways fast enough and the ground curves away under you. **Where you end up** depends on how high you go: low is fast and sharp but sees little, high is slow and sees everything, and one exact height hangs still. **Describing** any orbit takes six numbers, which are really six ways to describe a hula hoop. **Staying alive** up there means doing everything the air used to do for you, because the vacuum does none of it, and the thin air that's left eats your plastic. **Going somewhere else** is a fuel bill: fast is expensive, the lazy half-ellipse is cheap, and the only free lunch is bouncing off a moving planet.



orbit = fall + miss	LEO · MEO · GEO	a e i Ω ω ν	vacuum, outgassing	layers, atomic O	Lambert, Hohmann	flyby, Rosetta
S1 · S3	S1 · S3	S3	S2	S2	S1 · S3	S1 · S3

The 7 minutes at a glance

0:00–0:45	0:45–1:45	1:45–2:45	2:45–3:45	3:45–4:30	4:30–5:45	5:45–6:30	6:30–7:00
HOOK	SHELVES	HULA HOOP	FREE SERVICES	AIR BITES	FUEL BILL	FREE LUNCH	LANDING
falling & missing	LEO MEO GEO	6 numbers	vacuum	layers, atomic O	Hohmann	flyby	"nothing's free"

Pages 2–3: each scene like you're in 5th grade. Pages 4–10: the same scenes at engineer level, how it actually works, with numbers you can check, one scene per page. Page 11: the full 7-minute script. Pages 12–13: the three section talks with cue cards. Pages 14–15: quiz and answers. Page 16: how to rehearse.

Scenes 1–4: getting there and staying alive

🏀 Scene 1 · Falling and missing

Throw a ball sideways. It falls and lands. Throw it harder, it lands farther. Now throw it so hard that by the time it falls, the ground has curved away underneath it. It never lands. It just keeps falling *around* the Earth. That's an orbit. The space station isn't floating; it's falling, and missing, 16 times a day.

> **SAY: "An orbit is falling and missing. Go sideways fast enough and you never hit the ground."**

Draw: a mountain, three throws, the third one wrapping around.
Friend asks "so why do astronauts float?" → Because they're falling too, together with the station. Same as the drop on a roller coaster, but forever.

🏢 Scene 2 · Three shelves

Satellites live on three shelves. **Low shelf (LEO):** 200–2,000 km up, laps Earth every 90 minutes, sees a tiny patch in sharp detail. Space station, Starlink. **Middle shelf (MEO):** up to 36,000 km, 12-hour laps, GPS lives here. **Top shelf (GEO):** exactly 35,786 km, one lap takes exactly one day, so from the ground it hangs still. TV and weather satellites.
Higher = sees more, moves slower, sees less detail, and the radio takes longer. Low satellites only see one spot for minutes, so to cover the planet you fly a swarm: a constellation. GPS is a swarm of 24+; your phone needs 4 overhead.

> **SAY: "Higher means you see more but move slower. One exact height hangs still, and it has to be over the equator."**

Draw: Earth, three rings, a little flashlight cone from each to show footprint size.
Friend asks "why over the equator?" → Every orbit is centered on the middle of the Earth, and the equator is the only ground circle that matches. Park it over New York and New York slides away underneath.

🌀 Scene 3 · The hula hoop

Earth is a basketball; the orbit is a hula hoop around it. To tell someone exactly how the hoop sits you need six things: **how big** (a), **how squished** (e), **how tilted** (i), **which way the tilt faces** (Ω, swivel), **which way the squished end points** (ω, spin), and **where the ball is right now** (ν). Five freeze the hoop; one moves.
Most orbits are ovals with Earth off to one side, so there's a closest point (perigee) and a farthest point (apogee). Like a swing: fastest at the bottom (perigee), slowest at the top (apogee). And those are the two spots where you fire the engine: push at perigee and the far side rises; push at apogee and the near side rises.

> **SAY: "Six numbers describe any orbit: size, squash, tilt, swivel, spin, and where you are. Fastest at the low point, slowest at the high point."**

Draw: an egg with Earth off-center, two dots labeled fast and slow.
Friend asks "why six, not three?" → Size and shape don't tell you how the hoop is turned in space. That takes three more.

🧯 Scene 4 · Five free services

Down here, the air does five jobs you never notice: it's **sunscreen** (ozone blocks harsh UV), it's **pressure** (holds gas inside materials), it's a **damper** (a buzzing ruler stops because air steals its energy), it's a **fan** (hot air rises and carries heat away), and it's a **ground wire** (static leaks away). Vacuum cancels all five. So in space: UV snaps plastics, trapped gas pushes out and fogs your lenses (outgassing), vibrations never die, hot spots stay hot (only radiators work), and charge builds until it sparks. And it all starts on the ground: clean rooms, because one speck can kill a sensor.

> **SAY: "Earth's air is doing five jobs for you right now. Space does none of them, so the spacecraft has to do all five itself."**

Draw: a box labeled spacecraft, five arrows pointing in, each labeled with what's missing.
Friend asks "so why does the ISS have those big white panels?" → Radiators. It makes its own heat and there's no air to blow it away, so it glows it off into space.

Scenes 5–7: the air bites, the fuel bill, the free lunch

🌡 Scene 5 · The air bites back

The air is a layer cake. Bottom layer (troposphere, 0–11 km): weather, planes, gets colder going up. Next (stratosphere, to 50 km): gets *warmer*, because the ozone is here soaking up UV and turning it into heat. Next (mesosphere, to 90 km): coldest, meteors burn. Then the thermosphere (to 500 km): the thermometer says 1000 °C but there's almost no air, so nothing can hand that heat to you. The space station flies through it and doesn't melt. Temperature is how fast molecules move; heat is how many there are to hit you.

Up there UV also snaps oxygen pairs into single oxygen atoms, which are desperate to grab onto anything. The spacecraft plows into them at 7.5 km/s and they eat its plastics like sandpaper. Metal shrugs; Kapton film gets shredded. Fix: coatings.

What the air is made of, bottom to top: below 100 km it's a stirred mix, the same everywhere: nitrogen pairs 78%, oxygen pairs 21%, argon 0.9%, carbon dioxide 0.04%, a little water vapor, and ozone (three oxygen atoms) as a few parts per million in the stratosphere. Above 100 km the stirring stops and gases sort by weight: single oxygen atoms rule 200–600 km (LEO), helium 600–1,500 km, and hydrogen on top, so thin it leaks into space. The ionosphere (60–1,000 km) overlaps all of it: sunlight knocks electrons off atoms, and the charged soup bends radio and delays GPS.

> **SAY: "The top of the atmosphere is 1000 degrees but can't burn you, because there's nothing there. What is there is loose oxygen atoms that eat plastic."**

Draw: a stack of five bands with a zigzag temperature line beside it.

Friend asks "1000 degrees and it doesn't melt?" → One sparkler is 1000 degrees and doesn't warm a room.

💸 Scene 6 · The fuel bill

First, the ladder of ways to visit a planet: fly past it, orbit it, drop a probe into its air, land on it, drive on it, or crash into it on purpose. Each rung costs more fuel to slow down and brings back more science.

Now, getting there. You don't aim at Mars; you aim at where Mars will be. The cheapest route is a lazy half-ellipse that drifts outward and meets Mars halfway around the Sun: the **Hohmann transfer**, about 8.5 months, two engine burns. You could go faster, but every bit faster costs wildly more fuel, and past a point the rocket can't be built. **Lambert's problem** is the calculator: two points plus a time, out comes the orbit and the fuel bill. And you can only launch when Earth and Mars line up, every 26

🚚 Scene 7 · The free lunch

For really far places the lazy throw is still too expensive. So you cheat. Swing your spacecraft close past a planet. In the planet's eyes you come in and go out at the same speed, just bent. But the planet is moving around the Sun, and the bend lines you up to carry some of that motion along. You leave faster than you arrived, for free. That's a **flyby** or **gravity assist**: a ball bouncing off the front of a moving truck. The planet pays for it by slowing down an amount nobody could ever measure.

More speed means a bigger orbit, which means you can reach farther. Pass in front of the planet instead of behind and you slow down, which is how you brake at Mercury. The price is time: Rosetta took 10 years and four flybys to reach a comet. New Horizons took 9.5 years to Pluto with one Jupiter boost, and arrived too fast to ever stop, which is why it was a flyby.

> **SAY: "Bounce off a moving planet and you leave faster than you came. It's free, but it takes years."**

Draw: a planet with a thick arrow showing it moving, a dashed path bending around it.

Friend asks "isn't that free energy?" → No, it comes from the planet. Jupiter slowed down by about a trillionth of nothing.

🎯 The landing line (end on this)

"So that's the whole thing. Space gives you nothing for free. Getting there is falling and missing. Staying alive means doing the air's five jobs yourself. And going anywhere is a trade between time and fuel, unless you can borrow a planet. Every spacecraft ever built is just a different answer to those trades."

> **Then stop talking. Let them ask.**

🔁 Where this came from

Scenes 1, 2, 6, 7 = Section 1 + Section 3. Scenes 4, 5 = Section 2. Scene 3 = Section 3. Your notebook already has every fact on this page; what's new here is the order and the connecting sentences.

Scene 1 · Falling & missing: How it actually stays up (no engines involved)

A spacecraft in orbit does nothing to stay in orbit. It doesn't fire sideways; it doesn't fire at all. Gravity is the only force on it, and gravity is what keeps it there. Here is the arithmetic you can check yourself, for the ISS at 400 km.

Step	In short	How it actually works	Official name / why it matters
1 · The rocket's job	give it sideways speed, once	Launch looks vertical for a minute, then tips over. Most of the burn is spent going sideways. At engine cutoff the spacecraft is moving horizontally at ~7.7 km/s. Height is cheap; sideways speed is the expensive part.	Official name: orbital velocity (circular velocity), $v = \sqrt{(\mu/r)}$.
2 · The fall	gravity pulls it down ~4.3 m every second	Gravity at 400 km is 8.7 m/s², about 90% of surface gravity. In one second the spacecraft drops $\frac{1}{2} \cdot 8.7 \cdot 1^2 \approx 4.3$ m toward Earth.	There is plenty of gravity up there. 'No gravity in space' is false.
3 · The miss	the ground curves away ~4.3 m every second	In that same second it moves 7.7 km sideways. Over 7.7 km, Earth's surface curves away by $d^2/(2R) = (7{,}670\text{ m})^2 / (2 \times 6{,}771{,}000\text{ m}) \approx 4.3$ m. Drop equals curve. Distance to the ground never changes.	That cancellation IS the orbit. Falling and missing is arithmetic, not a metaphor.
4 · Why gravity is the string	gravity = centripetal force	Swing a ball on a string: the string's inward pull is what bends a straight path into a circle. Gravity is Earth's string. A circular orbit is exactly the speed where gravity's pull equals the inward pull that circle needs.	Official name: centripetal force. For a satellite, gravity provides it.
5 · Wrong speed?	no engine needed for any of these	Too slow for the height: drops more than the ground curves, path bends into the atmosphere. Too fast: climbs, slows, comes back round as an ellipse. Faster than escape speed ($\sqrt{2} \times$ circular): never comes back.	Different starting speeds, same physics. See vis-viva.
6 · What engines DO run for	station-keeping and attitude, not 'staying up'	Station-keeping: undoing tiny losses. ISS loses ~2 km/month to thin air and reboosts for a few minutes about monthly. GEO sats do small burns every few weeks against the Moon's and Sun's tug to hold their slot. Attitude control: pointing, via reaction wheels; it never changes the orbit. A tumbling spacecraft is still in a perfect orbit.	Two separate jobs, two separate names: orbit maintenance (station-keeping, reboost) vs attitude control (ADCS).

- The sentence to own: it doesn't stay up by pushing. It stays up by moving sideways so fast it keeps missing the ground. Engines only run to fix pointing or to undo tiny drag losses.
- Why 'weightless': you feel weight only when something pushes back on you (the floor). In free fall nothing does, so you float, even though gravity is 90% of normal.
- A GEO satellite's lifetime is set by its station-keeping fuel. When that runs out it still orbits fine; it just drifts out of its parking slot and is moved to a graveyard orbit.

Scene 2 · Three shelves: Why higher is slower, how a satellite gets to GEO, and how GPS actually finds you

Three mechanisms hide behind the three shelves. Each one has numbers you can check.

Step	In short	How it actually works	Official name / why it matters
1 · Why higher = slower	gravity weakens with distance (inverse square)	Gravity falls off as $1/r^2$. At LEO (r = 6,771 km) you need 7.67 km/s sideways to balance it; at GPS height (26,560 km) only 3.87 km/s; at GEO (42,164 km) only 3.07 km/s. Slower speed AND a bigger circle, so the lap time balloons: 92 min → 12 h → 24 h.	$v = \sqrt{(\mu/r)}$ and $T = 2\pi\sqrt{(r^3/\mu)}$. Period grows with r^1.5 (Kepler's third law).
2 · Why 35,786 km exactly	solve T = one sidereal day for r	Set T = 86,164 s (23 h 56 m 4 s) in $T = 2\pi\sqrt{(r^3/\mu)}$ with $\mu = 398{,}600$ km³/s² and solve: r = 42,164 km from the center. Subtract Earth's 6,378 km equatorial radius: 35,786 km altitude. Nobody chose it; the equation did.	Any other altitude gives a different period, so the satellite drifts east or west across the sky.
3 · How a satellite reaches GEO	parking orbit → GTO → apogee kick	Rocket drops it in a ~200–300 km parking orbit (7.8 km/s). Burn 1 at perigee, +2.4 km/s, stretches the orbit into a Geostationary Transfer Orbit with apogee at 35,786 km. Coast 5.3 hours to apogee (now moving only 1.6 km/s). Burn 2 there, +1.5 km/s, rounds it off at 3.07 km/s. Total ~3.9 km/s after launch.	It's your perigee/apogee burn rule, applied. A launch from 28.5° latitude (Florida) also needs a plane change to reach i = 0°, which is why GEO launches from near the equator (Kourou, 5°) are cheaper.
4 · How GPS actually finds you	time stamps → distances → spheres	Each GPS satellite broadcasts the exact time and its own orbit. Your phone compares the time it hears with its own clock: the delay × the speed of light = distance to that satellite. One distance puts you on a sphere; three spheres cross at a point. The fourth satellite corrects your phone's cheap clock, which is wrong by far more than a few meters' worth of light-time.	Light travels 30 cm per nanosecond, so GPS needs clocks good to nanoseconds. The satellites carry atomic clocks; your phone doesn't, hence the fourth satellite.
5 · Why MEO for GPS	high enough to see a lot, low enough to be strong and fast	At 20,200 km each satellite sees ~38% of Earth, so 24 satellites in 6 planes guarantee 4+ overhead anywhere. GEO would need fewer but the signal would be weaker and the poles uncovered; LEO would need hundreds.	Orbit choice is a trade-off between coverage per satellite and signal strength/delay.
6 · Holding a GEO slot	station-keeping box ±0.05°	The Moon and Sun tug a GEO satellite north-south about 0.85° per year; Earth's slightly lumpy equator drifts it east-west. Small burns every few weeks, ~50 m/s of Δv per year, keep it in its box. When the fuel's gone, a last burn lifts it ~300 km into a graveyard orbit.	A typical GEO satellite carries ~15 years of station-keeping propellant; that, not the electronics, ends its life.

- The one rule behind the three shelves: gravity weakens with distance, so higher orbits are slower, longer, and hang still only at the single height where one lap equals one day.
- Why launch sites matter: Earth's spin gives a free 0.46 km/s eastward at the equator, zero at the poles. Launching east, near the equator, is the cheapest way up.
- GPS gives position; it does not track you. Your phone only listens. Nothing goes up.

Scene 3 · The hula hoop: How a burn changes the six numbers, and where the numbers come from

The Classical Orbital Elements aren't just labels; each engine burn moves specific ones and leaves the rest alone. That's how you steer. And every set of elements you'll ever see online was measured, not assumed.

Step	In short	How it actually works	Official name / why it matters
1 · Prograde burn at perigee	raises apogee → a ↑, e ↑	Speed up at the low point and the far side of the orbit rises. Size (a) grows, the ellipse stretches (e grows). Tilt (i), swivel (Ω) and spin (ω) don't change, because you pushed along the direction you were already moving, inside the plane, at the perigee itself.	This is burn 1 of a Hohmann / GTO.
2 · Prograde burn at apogee	raises perigee → a ↑, e ↓	Speed up at the high point and the near side rises; the ellipse rounds out toward a circle. Same three untouched: i, Ω, ω.	This is burn 2, the circularization or apogee kick.
3 · Retrograde burns	the mirror image	Slow down at apogee and perigee drops (how you de-orbit: a retrograde burn at the high point dips the low point into the atmosphere). Slow down at perigee and apogee drops.	De-orbit burns for returning capsules are retrograde, roughly 100 m/s from the ISS.
4 · Sideways (out-of-plane) burn at a node	changes i	Push perpendicular to the orbital plane while crossing the equator and the tilt changes. Cost: $\Delta v = 2 \cdot v \cdot \sin(\Delta i/2)$. At LEO speed, 1° of tilt costs ~134 m/s; 28.5° to 0° costs ~3.8 km/s, more than reaching orbit. So you change planes at apogee, where v is smallest, or launch into the right plane to begin with.	Why inclination is the expensive element and GEO launches happen near the equator.
5 · Out-of-plane burn away from a node	changes Ω (and i)	The same sideways push done elsewhere on the orbit swivels the plane around Earth's axis instead of (or as well as) tilting it. Rarely done on purpose; Ω is usually left to drift.	Nature changes Ω for free via Earth's bulge, see row 7.
6 · Radial burn	rotates ω	Push straight toward or away from Earth and the line of apsides rotates: perigee moves around the orbit without the size changing much. Used to re-point an elliptical orbit's low point over a target.	Rare in practice; expensive for what it does.
7 · Where the numbers come from	state vector → elements, at an epoch	Radar, laser ranging or onboard GPS measure the satellite's position and velocity (six numbers: x, y, z, vx, vy, vz, called the state vector) at one instant (the epoch). Pure geometry converts that into a, e, i, Ω, ω, ν. The public format is the Two-Line Element set (TLE), which is why every element comes with an epoch date.	Elements are a different way of writing position-and-velocity, chosen because five of them barely change.
8 · Why elements drift	J_2 and friends	Earth isn't a perfect sphere; its equatorial bulge (the J_2 term) makes an inclined orbit's Ω swivel slowly westward (~5° per day for the ISS) and ω rotate within the plane. Engineers use it: pick i ≈ 98° and Ω swivels exactly once a year, giving a sun-synchronous orbit.	Perturbations aren't just noise; the right one is a free engine.

- Steering summary: speed along your motion changes size and shape; pushing sideways changes tilt and swivel; pushing toward Earth rotates the ellipse. Six numbers, three kinds of push.
- Why 'burn at the low point': the same Δv gives a bigger energy change where you're already moving fastest (the Oberth effect). Perigee burns are the most efficient place to spend fuel.
- Every orbit-change plan you'll ever read is a list of which element each burn moves.

Scene 4 · Five free services: How heat, gas and charge actually behave in vacuum, with numbers

The five free services are the picture. These are the mechanisms underneath, the ones the engineers design to.

Step	In short	How it actually works	Official name / why it matters
1 · How a radiator works	power radiated ∝ T^4 (Stefan–Boltzmann)	Every surface glows heat as infrared light at a rate $\sigma \cdot \varepsilon \cdot T^4$ per square meter. A 1 m² panel at 300 K (27 °C) with emissivity $\varepsilon \approx 0.9$ sheds about 410 W into cold space. Double the absolute temperature and you shed 16× as much. So radiators are kept hot and big, and painted with coatings that have high emissivity (radiate well) but low solar absorptivity (don't soak up sunlight): white paint, silvered Teflon.	The ISS radiators total ~1,700 m² to dump ~75 kW. No fan on Earth comes close for that area.
2 · How heat gets to the radiator	conduction, heat pipes, pumped loops	Metal straps conduct a little. Heat pipes do most of it: a sealed tube with a liquid (ammonia) that boils at the hot end, flows as vapor to the cold end, condenses, and wicks back. No pump, no power, moves kilowatts. Big stations use pumped ammonia loops.	Convection replaced by a plumbing trick that still works in free fall.
3 · Why hot and cold at once	sun side +120 °C, shade side −150 °C	Sunlight delivers 1,361 W/m². A surface facing the Sun with nothing cooling it settles where radiated = absorbed, which for a black surface is ~120 °C; a shaded surface radiating to 3 K space drops toward −150 °C. Spinning the spacecraft, MLI blankets and heaters keep electronics in their −20 to +50 °C window.	Hence 'barbecue roll' on Apollo: rotate slowly so no side bakes.
4 · How outgassing contaminates	hot source → cold sink	Trapped water, solvents and plasticizers diffuse out faster when warm. The vapor travels in straight lines (no air to scatter it) and sticks to whatever cold surface it hits. Optics and detectors are deliberately the coldest things on board, so they're the sink. A monolayer of film on a star tracker lens costs real pointing accuracy.	Standard test (ASTM E595): 125 °C in vacuum for 24 h; pass = TML < 1% and CVCM < 0.1%. Bake-out beforehand gets the loss out on the ground.
5 · How charging builds and arcs	photoelectric vs plasma collection	Sunlit surfaces lose electrons to UV (photoelectric effect) and go positive; shaded insulating surfaces collect plasma electrons and go negative. Two surfaces a few cm apart can end up kilovolts apart. When the gap breaks down, an arc jumps: a current spike that resets computers or punches through solar-array insulation.	Fix: every surface conductive (indium tin oxide over glass, carbon-loaded paints) and bonded to the structure with straps, so nothing can be more than a volt from anything else. GEO, in Earth's magnetotail during storms, is the worst place for it.
6 · Why vibration persists	damping ratio ~0.1–1%	A metal structure's own internal damping is tiny: a strut rung like a bell in vacuum loses only a fraction of a percent of its energy per cycle. Launch shakes everything hard (that's what vibration testing checks). In orbit the worries are reaction-wheel jitter and slewing: a big antenna swung quickly keeps wobbling for minutes, blurring images.	Hence dampers, slow slews, and 'settle time' built into every observation plan.

- Design rule: in vacuum you must create every heat path and every charge path on purpose. Nothing happens for free, and nothing happens by accident either, which is also why thermal and charging behavior is so predictable once modeled.
- The T^4 law is why a small temperature rise makes a radiator much more effective, and why cold things in space are very hard to cool further.
- A cheap sanity check for any spacecraft picture: the gold/silver crinkly stuff is MLI, the big white flat panels are radiators, the dark flat panels are solar arrays.

Scene 5 · The air bites back: What the air is made of, floor by floor

Think of the atmosphere as a smoothie at the bottom and a layered drink at the top. Below about 100 km everything is stirred together, so the recipe is the same everywhere. Above about 100 km the stirring stops and the gases sort themselves by weight: heavy ones low, light ones high. Sunlight also starts breaking molecules apart up there.

Height	Zone	What's there	Why it matters
0–100 km	the stirred mix (homosphere)	N_2 78% · O_2 21% · Ar 0.9% · CO_2 0.04% · water vapor 0–4% (mostly below 11 km) · trace neon, helium, methane · ozone O_3 is a few parts per million, peaking at 15–35 km	Same recipe at every height here. This is the air you breathe and the air planes fly in.
100–200 km	the breakup zone	N_2 still the main gas, but UV is now splitting O_2 into single O atoms; atomic O climbs fast	Where the atmosphere stops being 'air' as you know it. Meteors have already burned up below this.
200–600 km	atomic oxygen country	atomic O is the number-one gas; N_2 fading; a little N and He	LEO. The ISS, Hubble and Starlink fly here. Atomic O erodes polymers; what's left of the density causes drag.
600–1,500 km	helium shell	He takes over from O (the boundary moves up and down with the Sun's activity)	Too thin to matter much for drag; GPS and GEO are far above this.
1,500 km and up	hydrogen fringe (exosphere / geocorona)	mostly single H atoms, so thin they can escape Earth entirely	A faint hydrogen halo glows in UV out to tens of thousands of km.

- Why the order: nitrogen pairs and oxygen pairs are heavy, single oxygen atoms are half an O_2, helium is lighter still, hydrogen is the lightest thing there is. Once the stirring stops, each gas settles at the height its weight allows, like oil over water.
- Where ozone fits: ozone is just three oxygen atoms stuck together. It forms in the stratosphere when UV hits O_2 and the loose atoms grab onto other O_2. It's rare (a few molecules per million) but it absorbs so well that it blocks most harmful UV. Above the stratosphere there isn't enough air for it to form, so UV comes through raw.
- The ionosphere overlaps all of this, roughly 60–1,000 km: sunlight knocks electrons off atoms, leaving charged particles. It bends and reflects radio, which is why AM radio travels far at night and why GPS signals get slightly delayed. It also grows and shrinks between day and night.
- 'Anomalous O' on the lecture's chart: a small population of extra-hot oxygen atoms the model adds above ~300 km. Still oxygen; not a separate thing to memorize.
- Why a spacecraft cares, in one line each: atomic O eats plastics (LEO); the thin air causes drag that the Sun's activity can double or triple; the ionosphere messes with radio and GPS; helium and hydrogen are too thin to matter.

Scene 6 · The fuel bill: The Hohmann and rocket-equation numbers, worked

The lazy throw isn't just cheapest in spirit; here is what each burn costs and why fast transfers are impossible, with the actual arithmetic.

Step	In short	How it actually works	Official name / why it matters
1 · LEO → GEO Hohmann, worked	$\Delta v_1 = 2.42$, $\Delta v_2 = 1.47$, total 3.89 km/s	Circular speed at 300 km: √(398600/6671) = 7.73 km/s. Transfer ellipse: a = (6671 + 42164)/2 = 24,418 km. Vis-viva at perigee: √(398600·(2/6671 − 1/24418)) = 10.15 km/s, so burn 1 = 10.15 − 7.73 = 2.42. At apogee: 10.15 × 6671/42164 = 1.61 km/s. Circular speed at GEO = 3.07, so burn 2 = 3.07 − 1.61 = 1.47. Time of flight = $\pi\sqrt{(a^3/\mu)}$ = 5.3 h.	Every number here comes from two formulas you already have: vis-viva and circular speed.
2 · Earth → Mars Hohmann, worked	~3.6 km/s to leave, ~2 km/s to capture, 259 days	Around the Sun: Earth moves at 29.8 km/s, the transfer ellipse needs 32.7 at perihelion, so you must leave Earth's neighborhood with $v_\infty \approx 2.9$ km/s. But you start in a gravity well: from LEO the burn is $\sqrt{(v_{esc}^2 + v_\infty^2)} - v_{circ} = \sqrt{(11.0^2 + 2.9^2)} - 7.7 \approx 3.6$ km/s. You arrive at Mars with $v_\infty \approx 2.6$ km/s; braking into a loose orbit costs ~2 km/s more. Travel time: half the transfer ellipse's period ≈ 259 days.	Why the departure burn is only 3.6 and not 11 + 2.9: you're already doing 7.7 km/s in LEO, and escape speed is √2 × that.
3 · Why you can only go every 26 months	phase angle 44°	Mars must be ~44° ahead of Earth at departure so it arrives at the far end of the ellipse when you do. Earth laps the Sun in 365 days, Mars in 687; the relative angle repeats every 1/(1/365 − 1/687) = 780 days.	Miss the window by a month and the Δv climbs steeply; miss by six and it's not possible.
4 · Why 20 days is impossible	the rocket equation	Tsiolkovsky: $\Delta v = Isp \cdot g_0 \cdot \ln(m_0/m_f)$. A good hydrogen engine has Isp ≈ 450 s, so $Isp \cdot g_0 \approx 4.4$ km/s. For the 3.6 km/s Hohmann departure, $m_0/m_f = e^{(3.6/4.4)} \approx 2.3$: 56% of the departing mass is propellant, fine. A 20-day trip needs ~60 km/s: $e^{(60/4.4)} \approx 800{,}000$. The ship would need to be 99.9999% propellant. No structure can be that light.	Fuel cost grows exponentially with Δv. That single fact shapes every mission.
5 · How Lambert's problem is actually solved	one free parameter, one equation	Infinitely many ellipses with the Sun at a focus pass through two given points. They differ by their semi-major axis. Lambert proved the time of flight between the points depends only on a, the straight-line distance between them, and the sum of the two radii. So you pick the a whose time of flight matches your deadline (by iteration), and the departure and arrival velocities fall out, hence the Δv.	Mission designers run this for thousands of date pairs and plot the Δv as a contour map, the 'porkchop plot', to pick the launch day.
6 · Why burns happen at perigee	the Oberth effect	Kinetic energy is $\frac{1}{2}mv^2$, so adding a fixed Δv adds more energy when v is already large. A 1 km/s burn at 10 km/s adds 10.5 units of energy per kg; the same burn at 2 km/s adds only 2.5. Deep in a gravity well you're moving fastest, so that's where fuel buys the most.	Why interplanetary departures burn from LEO, and why flybys dip as low as they dare.

- Chain of reasoning to keep: vis-viva gives speeds at each end → differences give Δv → rocket equation turns Δv into propellant fraction → exponential growth makes fast trips impossible.
- GTO numbers (2.42 + 1.47) and Mars numbers (3.6 + 2) are worth memorizing; they anchor every other mission you'll read about.
- Isp (specific impulse, in seconds) is the engine's fuel efficiency: chemical 300–450 s, ion engines 3,000+ s but with tiny thrust. Coming in the propulsion section.

Scene 7 · The free lunch: The flyby, worked: vectors, turning angle, and where the energy goes

A gravity assist is two frames of reference and one addition. Here is the mechanism with numbers, using a Jupiter flyby.

Step	In short	How it actually works	Official name / why it matters
1 · In Jupiter's frame	speed in = speed out, path bends by δ	Far from Jupiter you approach at some relative speed v_∞. Gravity bends you around a hyperbola and you leave at the same v_∞, rotated by the turning angle δ. Energy relative to Jupiter is conserved: nothing is gained or lost in this frame.	Hyperbolic excess velocity v_∞; turning angle δ. The closer you pass, the bigger δ.
2 · How much it bends	$\sin(\delta/2) = 1 / (1 + r_p \cdot v_\infty^2 / \mu)$	r_p = closest approach distance, μ = Jupiter's gravitational parameter (1.27×10^8 km³/s²). Example: v_∞ = 10 km/s, r_p = 500,000 km → $r_p \cdot v_\infty^2/\mu = 0.39$ → $\sin(\delta/2) = 0.72$ → δ ≈ 92°. Pass closer or slower and δ grows toward 180°.	Big planet, close pass, slow approach = big bend. Mars bends you little; Jupiter bends you a lot.
3 · In the Sun's frame	add Jupiter's own velocity	Jupiter moves around the Sun at 13 km/s. Your Sun-frame velocity is (your velocity relative to Jupiter) + (Jupiter's velocity). Before the flyby the two vectors partly cancel; after the bend they partly add. The magnitude changes even though the relative speed didn't.	Same addition as walking on a moving train: your speed relative to the ground depends on which way you walk.
4 · Worked gain	up to ~2 × v_{planet} in the ideal case	In the extreme 180° bounce, approaching head-on at v_∞ against the planet's motion and leaving along it, the Sun-frame speed goes from ($v_{planet} - v_\infty$) to ($v_{planet} + v_\infty$): a gain of $2 \cdot v_\infty$, capped by geometry at about $2 \cdot v_{planet}$. Real flybys bend 60–150°. Voyager 2 picked up roughly 10 km/s of Sun-frame speed at Jupiter.	Why Jupiter is the slingshot of choice: it's massive (big δ) and fast (13 km/s to borrow from).
5 · Where the energy comes from	momentum conservation	The spacecraft's gain is the planet's loss: $m_{spacecraft} \times \Delta v_{spacecraft} = M_{planet} \times \Delta v_{planet}$. With a 1-ton probe gaining 10 km/s, Jupiter (1.9×10^{27} kg) slows by ~5×10^{-24} m/s. Not free energy; just a ridiculously lopsided trade.	Energy and momentum are both conserved. The planet pays an amount no instrument could ever measure.
6 · Behind vs in front	gain vs brake	Pass behind the planet (on the trailing side of its motion) and the bend points you along its velocity: you gain. Pass in front and the bend points you against it: you lose. MESSENGER used six flybys of Earth, Venus and Mercury to shed speed so it could be captured at Mercury; Parker Solar Probe uses Venus the same way to fall toward the Sun.	Same trick, opposite sign. Geometry decides.
7 · Why it costs years	you must meet the planet where it is	A gravity assist only works if the planet is at the right place when you pass, so trajectories string together windows: Earth→Venus→Earth→Jupiter (Cassini, 1997–2004). Each loop around the inner solar system is a year or more of coasting.	Time is the currency you spend instead of propellant.

- In Jupiter's eyes nothing changed but your direction. In the Sun's eyes your speed changed because Jupiter was moving. Both are true at once; that's what 'frame of reference' means.
- Design levers: approach speed, closest-approach distance, and which side you pass on. Those three set the bend and the sign of the gain.
- After the flyby you're on a new, bigger (or smaller) orbit around the Sun with a new semi-major axis; that's the 'more speed → bigger a' line from Section 3.

Minute by minute: key line, napkin drawing, hand-off

Not a script to memorize word for word. Memorize the bold key line for each block and the drawing. Everything else you say in your own words. Total: 420 seconds.

0:00 0:45 **HOOK.** "Did you know the space station is falling right now? It just keeps missing." Ball off a mountain, three throws. **Key line:** an orbit is falling and missing. *Hand-off:* "So how high you go changes everything…"

0:45 1:45 **THREE SHELVES.** Low shelf 90 min, sharp, tiny view. Middle shelf GPS 12 h. Top shelf one day, hangs still, 35,786 km, over the equator. Higher = sees more, slower, blurrier. Swarms for low orbit. **Key line:** one exact height hangs still. *Hand-off:* "But how do you even describe an orbit to someone?"

1:45 2:45 **HULA HOOP.** Six things: size, squish, tilt, swivel, spin, where the ball is. Egg with Earth off-center: fastest at the low point, slowest at the high point, fire the engine at those two spots. **Key line:** six numbers, five freeze the hoop, one moves. *Hand-off:* "Okay, you're up there. Now space tries to kill you."

2:45 3:45 **FIVE FREE SERVICES.** Sunscreen, pressure, damper, fan, ground wire. Vacuum cancels all five: UV snaps plastic, outgassing fogs lenses, vibrations never die, hot spots stay hot, static sparks. Clean rooms. **Key line:** the air does five jobs for you; space does none. *Hand-off:* "And the little bit of air that IS up there is worse."

3:45 4:30 **THE AIR BITES.** Layer cake, zigzag temperature. Top layer 1000 °C but empty: a sparkler. Loose oxygen atoms eat plastic at 7.5 km/s. **Key line:** hot but empty, and the oxygen is hungry. *Hand-off:* "Now let's leave Earth entirely."

4:30 5:45 **THE FUEL BILL.** Ladder: fly past, orbit, probe, land, drive, crash. Don't aim at Mars, aim where it'll be. Lazy half-ellipse, 8.5 months, two burns, every 26 months. Lambert = the "by Tuesday?" calculator. **Key line:** fast is expensive, slow is cheap. *Hand-off:* "Unless you cheat."

5:45 6:30 **THE FREE LUNCH.** Bounce off the front of a moving truck. Planet's view: same speed, bent. Sun's view: faster. Rosetta 10 years, New Horizons 9.5. Pass in front to brake. **Key line:** free speed, but it costs years. *Hand-off:* "So here's the whole thing in one breath."

6:30 7:00 **LANDING.** "Space gives you nothing for free. Getting there is falling and missing. Staying alive is doing the air's five jobs yourself. Going anywhere is time versus fuel, unless you borrow a planet." Then stop and let them ask.

⚡ Running long? Cut Scene 5 (the air) to one sentence: "and the thin air that's left eats plastic." Running short? Add the ISS-radiator question to Scene 4. Never cut the hook or the landing.

Practice in pieces: Section 1 and Section 2

Each section is its own short talk with its own opener, blocks, landing line and cue card. Get each one to a clean run 3 before attempting the full 7 minutes. These match the Teach tab in the app.

🛰️ Section 1 · Intro & mission design · 3¼ min

Friend opens with: “So what even is a satellite, and why are there different kinds?”

0:00 0:45 **HOOK.** An orbit is falling and missing. *Draw:* ball off a mountain, three throws. *Hand-off:* “So once something's up there, what is it FOR?”

0:45 1:30 **FOUR JOBS.** Every satellite looks out, looks down, relays, or carries something alive. *Draw:* four boxes: telescope, camera pointing down, radio dish, a little dog. *Hand-off:* “And where it lives depends on the job…”

1:30 2:30 **THREE SHELVES.** Low is fast and sharp, high sees everything, one exact height hangs still over the equator. *Draw:* Earth, three rings, flashlight cones. *Hand-off:* “Now leave Earth. How do you visit another planet?”

2:30 3:15 **THE LADDER.** Fly past, orbit, probe, land, drive, crash. Each rung costs more fuel and gives more science. *Draw:* a planet with six paths, numbered.

Landing line: “Every spacecraft is a different answer to one question: how close do you dare get, and what will it cost?”

1 · FALL & MISS — ball off a mountain
2 · 4 JOBS — look out / look down / relay / live
3 · 3 SHELVES — 90 min / 12 h / 1 day · 35,786 · equator · swarm
4 · LADDER — flyby → orbiter → probe → lander → rover → penetrator
LAND — “every rung costs more and gives more”

🧯 Section 2 · Space environment · 3 min

Friend opens with: “What's actually so dangerous about space? There's nothing there.”

0:00 1:15 **FIVE FREE SERVICES.** The air does five jobs for you: sunscreen, pressure, damper, fan, ground wire. Space does none. *Draw:* box with five arrows in, each labeled. *Hand-off:* “And the little bit of air that IS up there is worse.”

1:15 2:15 **AIR BITES BACK.** The top of the atmosphere is 1000 degrees but empty, and the loose oxygen atoms eat plastic. *Draw:* layer cake with a zigzag temperature line. *Hand-off:* “So what's actually floating around up there?”

2:15 3:00 **WHAT'S IN THE AIR.** Stirred nitrogen and oxygen near the ground, then single oxygen, then helium, then hydrogen on top. *Draw:* five bands labeled with the gas that rules each.

Landing line: “Space isn't dangerous because something's there. It's dangerous because nothing is, and you have to bring all five of the air's jobs with you.”

1 · 5 FREE JOBS — sunscreen pressure damper fan ground wire · clean room
2 · AIR BITES — layer cake zigzag · 1000° sparkler · hungry oxygen 7.5 km/s
3 · WHAT'S IN IT — N_2/O_2 → atomic O (200–600) → He → H · ozone · ionosphere
LAND — “space does nothing for you; build all five”

Practice in pieces: Section 3

The longest of the three, because it carries the two ideas people actually ask about: how do you steer, and how do you get to Mars.

🌀 Section 3 · Orbital mechanics · 4¼ min

Friend opens with: “How do you even steer something in space? Isn't it just floating?”

0:00 1:00 **HULA HOOP.** Six numbers describe any orbit: size, squash, tilt, swivel, spin, where you are. Fastest low, slowest high. *Draw:* egg with Earth off-center, then a tilted hoop over a flat disk. *Hand-off:* “Pick the right hoop and you get the three shelves…”

1:00 1:45 **EARTH ORBITS.** Higher sees more but moves slower; 35,786 km hangs still; low orbits need swarms. *Draw:* three rings with periods written on them. *Hand-off:* “Now leave Earth entirely.”

1:45 3:00 **THE FUEL BILL.** You aim at where Mars will be. Fast is expensive, the lazy half-ellipse is cheap, every 26 months. *Draw:* two circles, dashed half-oval, a burn at each end. *Hand-off:* “Unless you cheat.”

3:00 3:45 **THE FREE LUNCH.** Bounce off a moving planet and you leave faster than you came. Free, but it costs years. *Draw:* planet with a thick arrow, dashed path bending around it. *Hand-off:* “So here's the whole thing in one breath.”

3:45 4:15 **LANDING.** Going anywhere is a trade between time and fuel, unless you can borrow a planet. Then stop.

🃏 Section 3 cue card

1 · HULA HOOP — size squash tilt swivel spin bead · fast low, slow high · burn there
2 · 3 SHELVES — 90 min / 12 h / 1 day · 35,786 · equator · swarm
3 · FUEL BILL — lazy throw 8.5 mo · every 26 mo · Lambert = by-Tuesday calculator · fast = $$$
4 · FREE LUNCH — moving truck · Rosetta 10 yr · NH 9.5 yr
LAND — “time vs fuel, unless you borrow a planet”

🧭 How the three talks become the full one

- The full 7-minute talk (page 4) is Section 1's hook, Section 3's hula hoop and shelves, Section 2's two blocks, then Section 3's fuel bill and free lunch, with one landing line for all of it.
- Section 1's "four jobs" and "the ladder" blocks are cut from the full talk to make time. Bring them back if a friend asks "what kinds are there?"
- Order of work: Section 1 to a clean run 3 → Section 2 → Section 3 → full talk.

⚡ On the call: keep this page and page 5 open. Read the key line, say it in your own words, draw the napkin picture, use the hand-off to move on.

Do it cold. Then check page 6.

Three kinds of question, mixed: facts (F), explain-why (W), and "a friend asks" (A). The W and A ones are the ones that matter for the 7 minutes. Score yourself honestly.

1. (W) In one sentence, why doesn't an orbiting satellite fall down?
2. (F) LEO, MEO, GEO: altitude band and lap time for each.
3. (W) Why is GEO at 35,786 km and not a round number?
4. (A) "Why can't a TV satellite hang over New York?"
5. (F) Geosynchronous vs geostationary: which one hangs over one spot?
6. (W) Why does GPS need 24+ satellites when a TV satellite needs 1?
7. (F) Name the six orbital parameters and what each one answers in one word.
8. (W) Where is a satellite fastest, and which two spots do you fire the engine at? What does each burn do?
9. (F) a = ? (formula, and measured from where?)
10. (F) e = 0 means? e near 1? e over 1?
11. (A) "What does the air do for us that space doesn't?" Name all five.
12. (W) Why is a vibration a bigger problem on a spacecraft than on a car?
13. (F) What is outgassing, and why keep outgassing materials away from sensors?
14. (A) "The thermosphere is 1000 °C. Why doesn't the ISS cook?"
15. (W) Why does the stratosphere get warmer as you go up?
16. (F) What is atomic oxygen, where does it come from, and where is it the main gas?
17. (W) Who's moving at 7.5 km/s in an atomic-oxygen hit, and why does it matter?
18. (F) Rank the six "visit a planet" spacecraft types from least to most committed.
19. (W) Why can't we fly to Mars in 20 days even though an orbit for it exists?
20. (F) Describe a Hohmann transfer: shape, angle, number of burns, time to Mars.
21. (F) Lambert's problem: what goes in, what comes out?
22. (A) "Isn't a gravity assist free energy?"
23. (W) In the planet's frame, what changes during a flyby? In the Sun's frame?
24. (A) "Why did Rosetta take 10 years?" and "Why couldn't New Horizons stop at Pluto?"

Scoring: 20+ solid → you're ready to rehearse. 15–19 → rehearse, but redo the missed ones first. Under 15 → one more review pass on the weak scene before you rehearse. Every miss becomes a question-bank card.

Answers, and which scene to review if you missed it

1. It is falling; it's just moving sideways so fast the ground curves away as fast as it falls. *(Scene 1)*
2. LEO 200–2,000 km, 1.5–2 h. MEO 2,000–35,786 km, 2–24 h (GPS 12 h). GEO 35,786 km, 24 h. *(Scene 2)*
3. It's the one altitude where one lap takes exactly one Earth rotation. Calculated, not chosen. *(Scene 2)*
4. Every orbit is centered on Earth's center; only the equator moves in a circle an orbit can match. Over New York it would drift north and south daily. *(Scene 2)*
5. Geostationary (one-day lap AND over the equator). Geosynchronous = one-day lap at any tilt. *(Scene 2)*
6. GPS is low (MEO), so each satellite sees a limited patch and moves; you need many so 4 are always overhead. A GEO satellite sees a third of Earth and never moves. *(Scene 2)*
7. a size, e shape, i tilt, Ω swivel, ω spin, ν position. *(Scene 3)*
8. Fastest at perigee. Burn at perigee → apogee rises; burn at apogee → perigee rises. *(Scene 3)*
9. a = (r_perigee + r_apogee) ÷ 2, both measured from Earth's center. *(Scene 3)*
10. 0 = circle. Near 1 = long cigar. Over 1 = open path that never closes (a flyby). *(Scene 3)*
11. Sunscreen (ozone vs UV), pressure, damper, fan (convection), ground wire (discharge). *(Scene 4)*
12. On Earth the air absorbs the vibration and it dies out. In vacuum nothing does; the spacecraft must damp it itself. *(Scene 4)*
13. Trapped gas leaking out of materials in vacuum. It condenses on cold surfaces, usually lenses and sensors, and fogs them. *(Scene 4)*
14. Almost no molecules up there. Temperature is molecule speed; heat needs numbers. A sparkler is 1000 °C and won't warm a room. *(Scene 5)*
15. The ozone layer lives there and absorbs UV; absorbed light becomes heat. *(Scene 5)*
16. Single O atoms made when UV splits O_2. Main gas from ~200–600 km, exactly where LEO satellites fly. *(Scene 5)*
17. The spacecraft. The atoms are nearly still; the spacecraft slams into them, so each hit carries enough energy to break a chemical bond. *(Scene 5)*
18. Flyby → orbiter → atmospheric probe → lander → rover → penetrator. Each step costs more fuel/risk, returns more science. *(Scene 6)*
19. The Δv needed is absurd; no rocket could carry the fuel. Fast is expensive. *(Scene 6)*
20. Half an ellipse touching both orbits; depart and arrive 180° apart; two burns; ~8.5 months to Mars. *(Scene 6)*
21. In: two positions + a travel time. Out: the one orbit connecting them, and therefore the Δv. *(Scene 6)*
22. No. The energy comes from the planet, which slows down by an unmeasurably tiny amount. *(Scene 7)*
23. Planet's frame: speed unchanged, direction bends. Sun's frame: speed changes because you add the planet's own motion. *(Scene 7)*
24. Rosetta used four flybys to build speed for free instead of carrying fuel; time was the price. New Horizons left Earth so fast (plus a Jupiter boost) that no fuel could brake it into Pluto orbit. *(Scene 7)*

Three runs, then do it for real

🎬 Three rehearsal runs

- **Run 1, with page 4 open.** Talk through all seven blocks, reading the key lines. Don't time it. Goal: feel the order.
- **Run 2, with only the cue card** (right). Phone timer running. Expect to run long. Note which block ate the time.
- **Run 3, nothing in front of you, phone recording.** Timer running. Then listen back and mark: every "um," every sentence that ended in a word you didn't explain, every place you reached for a fact and it wasn't there. Those three things are your fix list.
- **Then send me the recording or a transcript.** I'll grade it the way the friend would hear it, not the way a textbook would.

🧯 When you get stuck mid-sentence

- Pivot to the picture: "The way I think about it is…" then the analogy for that block (ball off a mountain, three shelves, hula hoop, five jobs, sparkler, lazy throw, moving truck).
- Forgot a number? Say the idea instead. "About a day" beats a frozen pause over "23 hours 56 minutes."
- Lost your place? Go to the hand-off line of the block you're in. They all start with "So…" or "But…"

📏 What "holding your own" sounds like

- You use the analogy first and the real word second: "it's like a hula hoop around a basketball, the six orbital parameters…"
- You can answer one "why" past each fact. Not two. One is enough for a friend.
- You say "I'm not sure, but I think…" for the things you're not sure about. That's what people who actually know things sound like.

🃏 Cue card (copy onto one index card)

1 · FALL & MISS — ball off a mountain
2 · 3 SHELVES — 90 min / 12 h / 1 day · 35,786 · equator · swarm
3 · HULA HOOP — size squash tilt swivel spin bead · fast low, slow high · burn there
4 · 5 FREE JOBS — sunscreen pressure damper fan ground wire · clean room
5 · AIR BITES — layer cake zigzag · 1000° sparkler · hungry oxygen 7.5 km/s
6 · FUEL BILL — ladder · lazy throw 8.5 mo · every 26 mo · fast = $$$
7 · FREE LUNCH — moving truck · Rosetta 10 yr · NH 9.5 yr
LAND — "space gives you nothing for free"

Eight lines. If you can talk for 7 minutes from these eight lines, you own it.

⚡ Order of work today: read pages 2–3 once → do the quiz cold (page 5) → check (page 6) → fix the misses → Run 1 → Run 2 → Run 3 → send me the recording.